

AKBAR SHAIK

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Summary

- **AI/ML Engineer with 4+ years** of experience delivering end-to-end machine learning solutions, specializing in **Large Language Models (LLMs), MLOps, computer vision, NLP, and data engineering**.
- Proficient in **Python, SQL, Bash**, and building production-grade ML pipelines for batch, streaming, and real-time inference across diverse business domains.
- Expertise in **Retrieval-Augmented Generation (RAG)** systems, **semantic search**, and vector databases (**FAISS, ChromaDB, Milvus**) for contextual, document-grounded, and multi-modal AI applications.
- Skilled in fine-tuning LLMs using **LoRA, PEFT**, instruction tuning, model distillation, and quantization to optimize latency, cost, and accuracy.
- Experienced in distributed training with **PyTorch DDP, Horovod, and DeepSpeed** for large-scale and multi-node model training.
- Designed scalable ETL pipelines using **Apache Spark, PySpark, Airflow, Kafka, Delta Lake, and Snowflake**, ensuring high data quality with Great Expectations and data lineage tracking.
- Strong **MLOps** expertise with **MLflow, Kubeflow, DVC, SageMaker Pipelines**, and **CI/CD** automation using **GitHub Actions** and **Jenkins** for reproducible, governed ML lifecycle management.
- Hands-on deployment experience using **Flask, FastAPI, TorchServe, Docker, Kubernetes, Helm**, and cloud-native ML platforms (**AWS SageMaker, Azure ML, GCP Vertex AI**).
- Adept at model monitoring, explainability, and governance with **SHAP, LIME, Captum, drift detection (PSI, KS Test), bias testing, model fairness metrics, and Responsible AI compliance**.
- Proven track record of delivering measurable business impact by improving operational efficiency, user satisfaction, and AI-driven decision-making through innovative ML solutions.

Technical Skills

- **Programming & Scripting:** Python, SQL, Bash, Git, REST APIs, GraphQL, Linux, Jupyter, VS Code
- **ML & DL Frameworks:** scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, CatBoost, ONNX, LoRA Fine-tuning, Model Distillation, Model Quantization, PyTorch Lightning, Distributed Training (Horovod, PyTorch DDP, DeepSpeed)
- **LLMs & NLP:** OpenAI GPT series, Hugging Face Transformers, BERT, T5, LLaMA, MPT, LangChain, Prompt Engineering, RAG, FAISS, ChromaDB, Milvus, Sentence-BERT, spaCy, NLTK, TextBlob, NER, Summarization, Q&A Systems, Contextual Search
- **Computer Vision:** OpenCV, YOLOv5, Detectron2, ResNet, Vision Transformers (ViT), Image Segmentation, Object Detection, OCR (Tesseract, EasyOCR)
- **Data Engineering & ETL:** Apache Spark, PySpark, Airflow, Kafka, Delta Lake, MongoDB, PostgreSQL, MySQL, AWS Glue, SQLAlchemy, Snowflake, BigQuery, Data Lakehouse
- **MLOps, Deployment & Orchestration:** Flask, FastAPI, TorchServe, Docker, Kubernetes, Helm, Terraform, SageMaker, Vertex AI, Azure ML, MLflow, DVC, Kubeflow, Model Registry, CI/CD (GitHub Actions, Jenkins)
- **Monitoring & Governance:** SHAP, LIME, Captum, Drift Detection (PSI, KS Test, ADWIN), Fairlearn, Model Cards, AI Governance, Great Expectations, Grafana, Prometheus, ELK Stack
- **Cloud & DevOps:** AWS (S3, SageMaker, Lambda, ECS, EKS, CloudWatch), Azure (ML Studio, Databricks), GCP (Vertex AI, BigQuery, Dataflow), Hugging Face Hub, GitOps
- **Visualization & Reporting:** Matplotlib, Seaborn, Plotly, Dash, Power BI, Tableau, Weights & Biases

Experience

Teledyne Technologies Inc

Jan 2024 – Present

Applied AI Engineer

- Designed a **RAG** pipeline using **LangChain, FAISS**, and **OpenAI GPT-4** for contextual document-grounded answers.
- Integrated **FastAPI, Docker**, and **AWS Lambda** to deploy low-latency LLM APIs for production inference.
- Developed custom embeddings for insurance FAQs and claims data using **HuggingFace Transformers** and **Sentence-BERT**.
- Created prompt templates with **PII masking** and **policy validation** to ensure secure and compliant conversations.
- Engineered a fallback intent classifier using **SVM (scikit-learn)** to divert high-risk inputs to human agents.
- Logged all model interactions in **MongoDB** and **AWS S3**, enabling auditable and retrainable inference pipelines.
- Tuned **BLEU/ROUGE** scores weekly via human-in-the-loop review to refine LLM response quality and structure.
- Achieved **42% faster responses** and **automated 60% Tier-1 queries**, improving **CSAT** scores with over 90% accuracy.
- Applied **guardrails** using policy rules and tracking to mitigate hallucinations and ensure regulatory compliance.

- Documented retrieval failures and updated vector indexes to resolve broken context flow in user interactions.
- Fine-tuned **GPT-based** models using **LoRA** and **parameter-efficient transfer learning** to improve domain adaptation speed by 35%.
- Implemented **real-time drift detection** on vector embeddings to maintain contextual accuracy in LLM retrieval.
- Orchestrated distributed model training using **PyTorch DDP** for high-volume insurance document classification tasks.
- Integrated **Captum**-based interpretability reports to provide explainability for regulatory audits.

Creative Newtech Ltd

Feb 2020 – Aug 2022

MLOps Engineer / ML Platform Engineer

- Built **MLflow**-integrated pipelines to track model versions, parameters, datasets, and training-to-deployment lineage.
- Automated **drift detection** using **Kolmogorov–Smirnov test** and **PSI**, alerting teams of distribution shifts.
- Integrated Great Expectations with ETL pipelines to validate schema, detect nulls, and enforce data quality.
- Enforced **CI/CD** workflows using **GitHub Actions** and **MLflow Registry** for controlled model promotion processes.
- Visualized training and production metrics using **Weights & Biases**, **Grafana**, and custom alerting dashboards.
- Generated **SHAP** and **LIME** explainability outputs for each model prediction reviewed by legal and business teams.
- Implemented responsible AI checks, covering bias tests, fairness metrics, and data consent validation.
- Automated model documentation with ethical concerns, assumptions, input features, and validation stats included.
- Reduced model failures by 70% through real-time monitoring, explainability, and traceable ML practices.
- Passed 2 regulatory audits by maintaining full model lineage, versioning, monitoring, and governance standards.
- Migrated legacy ML workloads to containerized, cloud-native deployments on **AWS SageMaker** and **Azure ML**.
- Implemented **PyTorch Lightning** for structured deep learning workflows, reducing training code complexity by 40%.
- Designed a feature store to centralize reusable features for fraud detection and demand forecasting models.
- Optimized inference pipelines with **ONNX Runtime**, cutting latency by 28% across production APIs.

Academic Projects

LLM-Powered Customer Support Agent

- Designed a RAG pipeline using LangChain and SwiftNLP, enabling semantic search across 100k+ user documents with Pinecone VectorDB for document retrieval.
- Implemented compliance-focused prompt engineering to ensure alignment with ethical AI guidelines and data privacy standards.

Fraud Detection System | PyTorch, HuggingFace, MLflow, AWS

- Built and deployed a real-time fraud detection pipeline using LLM ensembles, reducing model size by 60% through pruning and quantization.
- Developed a Kubernetes-based auto-scaling inference layer (500+ RPS) with GPU-aware scheduling and memory profiling for optimized performance.

E-Commerce Product Recommendation Classifier

- Created a collaborative filtering model (Python, TensorFlow) for customer transactions to deliver personalized product bundles.
- Increased the average order value by \$15 for a CPG client through targeted product recommendations.

Education

- **Master's Degree in Data Science**
University of North Texas
- **Bachelors of Technology**
Lakireddy Bailreddy College of Engineering

Certification

- AWS Certified Machine Learning – Specialty (IN-PROGRESS)
- Google Data Analytics Professional Certificate